

DLA Draft

Emergent Selective Learning in Fast/Slow Learners

Allocation-level selectivity of consolidation, without any selection objective

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Abstract

Selective learning—retaining some changes while discarding others—is usually treated as something an algorithm must explicitly implement (loss weighting, parameter gates, per-item replay). We ask whether it can instead *emerge* from an unselective learning rule, and whether the emergent selectivity is functionally real. We study a minimal “fast/slow” learner on a fixed Transformer backbone: a fast weight w_{fast} integrates the learner’s own gradient history (Adam-shaped, leaky), and at task boundaries an *unselective* rule copies a scaled w_{fast} into the slow weights at a uniform scalar coefficient—no per-parameter gate, no success signal, no alignment objective. On a 6.59M Chinese character GPT with two curriculum histories (easy→hard vs hard→easy) probed on a never-seen domain, we find: (1) history-dependent future adaptation is causally tied to the fast-weight trace (swap control, $n=12$); (2) the mechanism that transfers part of that trace into slow weights is the *direct* fast→slow write, not the architecture’s explicitly-designed success-gated eligibility trace Q , which is numerically ($\sim 0.4\%$ of write energy) and causally inert; and (3) the direct write’s **coordinate allocation is functionally necessary**: keeping every write’s energy identical but randomly permuting which coordinates receive it removes the entire effect ($n=12$, paired $t \approx 4.9$; shuffled allocation $\approx$ no

consolidation). Because the write is $\gamma \cdot W_{\text{fast}}$ with a single scalar coefficient, “where it writes” is fully self-organized from the fast trace’s own structure. We conclude that a minimal, unselective rule can exhibit *emergent, allocation-level selective learning*, that this selectivity carries causal weight for future adaptation, and that it coexists with an explicit but non-functional “selective” mechanism in the same system. In the same protocol, consolidation is not only a forward device: the sleep boundary itself protects earlier-task knowledge (no-sleep loses it, $n=12$, paired $t \approx 10$) and the direct write gives the best retention among consolidation variants. On a ten-task cross-domain curriculum ($n=8$), direct consolidation retains the earliest tasks best (paired $t \approx -3.6$ vs no-write; -10 vs no-sleep) while keeping forward learning at least as fast — an unselective fast $\rightarrow$ slow write behaves as a working continual-learning primitive, with no replay, gating, or importance machinery.

We report all claims with their evidence level and provide energy-matched controls throughout.

1. Introduction

1.1 The question: does selectivity need to be designed?

A learner with a memory faces a choice at every update: keep or discard. Machine learning usually makes this choice explicit—regularization that penalizes change to important weights (Kirkpatrick et al., 2017; Zenke et al., 2017), replay buffers that rehearse chosen items (Robins, 1995; De Lange et al., 2021), gated plasticity that opens or closes parameters (Miconi et al., 2019; Beaulieu et al., 2020), or curricula that choose what to learn next (Poirier & Silver, 2005; Bell & Lawrence, 2022; Li & Hiratani, 2025). Neuroscience likewise postulates specialized structures for selective retention (McClelland et al., 1995; Kumaran et al., 2016).

We ask a complementary, more mechanistic question:

If *no* selection signal is ever computed—no per-parameter gate, no success weighting, no importance measure—can selectivity still emerge, and if it does, is it functionally real or an epiphenomenon?

Selectivity is only meaningful if it has operational consequences. Our test is therefore causal: we must be able to *destroy the allocation* while keeping everything else (total write energy, magnitudes) identical, and show the outcome changes.

1.2 Why a fast/slow learner is the right object

Fast/slow weight architectures (Hinton & Plaut, 1987; Ba et al., 2016) are the simplest systems in which “what to consolidate” is a well-defined choice: a fast store accumulates recent changes, and a slow store must decide what to accept. Modern revivals (Schlag et al., 2021; von Oswald et al., 2023; Behrouz et al., 2025) and parameter-efficient additive methods (Houlsby et al., 2019; Hu et al., 2021; He et al., 2022) make the same conceptual split. These systems typically ship with an explicit consolidation policy. We strip the policy to its minimum—an unselective scalar-copy—and ask whether selectivity re-appears on its own.

1.3 The claim in one paragraph

In a minimal DLA (Section 3), two different curriculum histories produce different future adaptation on a never-seen domain, and this effect is carried by the fast-weight trace w_{fast} (Section 5.1). Consolidation into the slow store happens through two pathways: a *designed* one (a success-gated eligibility trace Q) and an *unselective* one (direct copy of w_{fast}). The designed pathway is inert; the unselective one carries the effect (Section 5.2). Critically, the unselective pathway is *selective in its allocation*: where it writes is dictated by the self-organized structure of w_{fast} , and destroying that allocation (energy-matched within-matrix shuffle) removes the effect (Section 5.3, $n=12$, paired $t \approx 4.9$). We call this **emergent selective learning**, and we are careful to define its level (allocation of writes, not experience-level or success-gated selection).

1.4 This is built on—not separate from—our prior fast/slow results

The learner we use is not new to this paper, and we deliberately build on the fast/slow results established earlier in the project rather than re-deriving them: – **Fast/slow separation protects consolidated memory**: in the same protocol, DLA matches a tuned AdamW on new-domain adaptation while slow-memory forgetting is ≈ 0 (-0.4%), i.e. the fast/slow split does not trade memory for plasticity. – **w_{fast} is a causal carrier of history-dependent future adaptation**: replacing a hard→easy learner’s fast weights with an easy→hard learner’s degrades future adaptation on a never-seen domain ($n=12$, norm/shuffle/module controls, direction replicated on a second backbone). – **Rule parameters ϕ and the success-gated eligibility trace Q do not carry that effect**; and standard

continual learners (AdamW, replay, EWC) in the same curriculum protocol are also history-sensitive, so history sensitivity is not unique to DLA—DLA is most robust after a difficult history.

These earlier findings are what make W_{fast} , Q and the slow store a *causally instrumented* setting rather than a toy: the question below—whether selectivity can emerge with no selection objective—is asked about states whose causal roles are already mapped.

2. Foundations we build on — and where we depart

This paper is written on top of four strands of prior work (each maps to a claim we make), and its contribution is the gap at their intersection: **does the selectivity that continual-learning, plasticity and consolidation research usually *injects* actually have to be injected?**

(1) Theoretical root: why fast/slow stores and consolidation exist.

Complementary learning systems theory explains memory/learning trade-offs by a fast, instance-based store and a slow, generalising store with offline consolidation (McClelland et al., 1995; Kumaran et al., 2016). DLA is a minimal weight-space instantiation of that split (§3.1), and our result speaks back to CLS: consolidation that is functionally *selective in its allocation* can arise without any gating/replay mechanism.

(2) Structural twin: frozen-base additive updates. Our effective weight $W_{\text{eff}} = W_{\text{slow}} + \text{softplus}(P) \cdot W_{\text{fast}}$ is the same additive decomposition used by parameter-efficient transfer — adapters (Houlsby et al., 2019) and LoRA (Hu et al., 2021), unified in (He et al., 2022) (§3.1). DLA differs in that the additive pathway is a shared, within-lifetime developmental state rather than a per-task fine-tune. Because of this structural twin, our central question — *whether “where the additive update lands” self-organizes and matters* — transfers directly to the LoRA/adapter setting in continual fine-tuning.

(3) Plasticity motivation. Plasticity is a fragile resource that ordinary continual training consumes (loss of plasticity; Dohare et al., 2024; Lyle et al., 2022; Nikishin et al., 2022). DLA’s per-parameter P is one response to that literature. Our finding offers a complementary view: keeping *beneficial structure in what is written to the slow store* may not require maintaining an explicit gate at all (§4.3–4.4).

(4) The nearest prior: task/curriculum ordering — where we explicitly differ. The order in which tasks are learned changes continual-learning outcomes, forgetting and consolidation of *seen* tasks (Bell & Lawrence, 2022; Li & Hiratani, 2025; Poirier & Silver, 2005; survey: Wang et al., 2022). We ask a related but distinct question (§1.1, §4.1): does order change *future adaptation to a never-seen domain*, and where inside the learner does the effect live? The difference is not cosmetic: because the test domain is never seen by either history, content-memory explanations of the order effect are excluded by design, which is what lets us localize the effect to w_{fast} and then to the *allocation* of the fast→slow write.

(5) Selection-by-design, and fast/slow as an engineering device. Continual-learning methods protect/replay/select explicitly (Kirkpatrick et al., 2017; Zenke et al., 2017; Robins, 1995; De Lange et al., 2021); plasticity methods gate per parameter, often meta-learned (Miconi et al., 2019; Beaulieu et al., 2020); fast/slow weight co-design appears as an optimization/RLHF engineering device (Ba et al., 2016; Qi et al., 2024) and in modern memory architectures (Schlag et al., 2021; von Oswald et al., 2023; Behrouz et al., 2025). These strands *inject* selection or use fast/slow for memory or efficiency; we instead hold selection absent and use the minimal fast/slow form as an instrument (within-individual development; cf. Andrychowicz et al., 2016).

Position in one sentence: existing work tells us history/order matters (4) and gives us the right architecture family (1, 2) and the warning that plasticity is fragile (3); we ask what happens when the selection machinery these strands normally provide is *not* provided — and find that selectivity still emerges, at the level of write allocation, with causal force.

3. System and definitions

3.1 The learner (minimal DLA)

We keep a standard Transformer backbone unchanged. Every wrapped weight matrix carries state:

$$w_{\text{eff}} = w_{\text{slow}} + \text{softplus}(P) * w_{\text{fast}}$$

This additive decomposition is the structural twin of frozen-base parameter-efficient methods — adapters (Houlsby et al., 2019) and LoRA (Hu et al., 2021; unified view: He et al., 2022) — with one difference that matters: here the additive pathway is a *within-lifetime developmental state* shared across tasks and updated by the learner’s own dynamics,

not a separately fine-tuned module. P is the per-parameter plasticity whose fragility under continual training is documented by loss-of-plasticity work (Dohare et al., 2024; Lyle et al., 2022; Nikishin et al., 2022). The fast/slow split itself follows complementary learning systems theory (McClelland et al., 1995; Kumaran et al., 2016), and fast/slow co-design has engineering precedent in optimization and RLHF (Qi et al., 2024) and modern memory architectures (Ba et al., 2016; Behrouz et al., 2025).

Wake step (per batch): backprop gives $g = dL/dW_{\text{eff}}$; Adam moments m, v are updated; the fast trace moves by

```
dw      = -eta_fast * softplus(P) * adam - fast_decay * W_fast
W_fast += dw
```

W_{fast} is therefore a *leaky integrator* (decay 0.02/step $\Rightarrow \approx 50$ -step horizon $\approx$ one curriculum stage) of the learner’s own Adam-shaped gradient history. P is a per-parameter positive scale updated by a progressxrelevance rule. A designed consolidation trace is also maintained: $Q \leftarrow (1-\alpha)Q + \alpha \cdot (dw \cdot \text{success})$, where success is a **single global scalar** derived from loss EMA—so the designed “selection” applies one coefficient to every parameter (this is deliberate; Section 3.3 formalizes why it cannot select).

Sleep (task boundary): two write pathways into W_{slow} :

```
W_slow += beta * Q + gamma * W_fast      # designed (Q) + unselective
direct copy
W_fast *= 0.5 ; Q *= 0.7 ; reset moments
```

The second term is the **unselective pathway**: every coordinate is copied with the same scalar γ . No objective, no gate, no importance weight—only W_{fast} ’s own structure decides, per coordinate, how much is written (the increment at coordinate i is $\gamma \cdot W_{\text{fast}_i}$).

3.2 Defining “selectivity” precisely (three levels)

level	definition	present in our system by design?
step/global	one scalar decides keep-vs-discard for the whole update	yes (success)
parameter	different coefficients per parameter based	no

level	definition	present in our system by design?
	on a per-parameter signal	
subspace	selection along learned/structured directions	no
allocation	an unselective coefficient, but the <i>magnitude/sign per coordinate</i> of what is written varies by coordinate	yes (emergent)

Our paper’s core construct is **allocation-level selectivity**: identical total energy and identical rule, but different coordinates receive different amounts because the source (w_{fast}) is structured. Whether this allocation *matters* is an empirical, causal question—answered by destroying it (energy-matched shuffle) while changing nothing else.

3.3 Protocol and measurement

- Two histories on the same three curriculum domains (Wikipedia / SFT-QA / science Wikipedia) in opposite orders: easy→hard (EH) and hard→easy (HE).
- Probe on a never-seen science slice D (40 steps, identical data/rule/budget/eval).
- Metrics: gain@40 (relative ppl improvement), LE_D, T80. All causal/ablation claims use **within-seed, within-arm paired contrasts** (the raw EH/HE gap is seed-noisy).
- Controls: cross-injection of state components; norm-matched / shuffled / module-localized swaps; consolidation ablations (Q-only / direct-only / none); and the energy-matched **allocation shuffle** (per-matrix random permutation of each sleep’s $\gamma \cdot w_{\text{fast}}$ increment).
- Every claim is tagged by evidence level (causal / controlled / correlational / negative). No pre-registration; $n=10 \rightarrow n=12$ transparent; no seed selection; no hyperparameter tuning for effect size.

Backbone: 6.59M Chinese character GPT (6 layers, 8 heads, 256 dim, vocab 7280) pretrained on Chinese Wikipedia. Second setting (replication in direction only): 10.65M Shakespeare char GPT, $n=2$.

4. Results

4.1 History lives in the fast trace (carrier, causal)

Replacing a hard→easy learner's w_{fast} with an easy→hard learner's significantly degrades future adaptation on D: paired gain@40 $\Delta \approx -0.019$, 95% CI $[-0.033, -0.005]$, sign $p=0.019$ (10/12 negative, $n=12$). Norm-matching does not rescue it ($d \approx -0.90$), within-matrix shuffling does not remove it ($d \approx -1.25$), and module-localized swaps implicate MLP ($d \approx -1.05$) and attention ($d \approx -1.06$) more than embedding ($d \approx -0.87$). The direction replicates on the Shakespeare backbone ($n=2$). Slow weights, plasticity and rule parameters transfer little alone (component decomposition).
Level: causal (single small backbone, $n=12$, controls + partial replication).

Relation to task/curriculum-order results. This mirrors, but differs from, the task-order results of continual learning (Bell & Lawrence, 2022; Li & Hiratani, 2025; Poirier & Silver, 2005): those measure how order changes *performance or forgetting on the seen curriculum*. Here the probe domain D is never seen by either history, so the effect must ride on the learner's internal state rather than on stored content about D — which is what allows the further steps below (carrier localization, then allocation shuffle). Where those works propose curricula as a *control* over outcomes, our claim is narrower and more mechanistic: the order effect is a *state effect*, and that state effect is itself selective in where it consolidates.

4.2 The directional trace is self-organized (no alignment objective exists)

A static code audit of the full wake/sleep/meta path confirms there is **no alignment operator anywhere** (no cosine/dot/objective between gradients and a history direction); directional content can only arise from w_{fast} 's leaky integration of Adam-shaped gradients. Empirically, the per-seed alignment $\cos(g_0, \Delta w_{\text{fast}})$ between the D-gradient and the same seed's history contrast predicts HE adaptation: $r=0.87$ (permutation $p \approx 0.000$), leave-one-seed-out $r \in [0.81, 0.91]$, survives FDR ($q \approx 0.003$), survives norm-confound controls (partial $r \approx 0.77$), and module-localizes to MLP/attention—the same modules implicated causally in §4.1. *Level: correlational, consistent with emergence; no direction-manipulation experiment.*

4.3 Of two consolidation pathways, only the unselective one works

Measured on saved bodies: $\|Q\| \approx 0.002$ vs $\|W_{\text{fast}}\| \approx 3.6$; the designed Q write is $\approx 0.4\%$ of the direct write. Consolidation ablation on the HE arm ($n=12$, gain@40):

variant	mean	paired t vs full	paired t vs direct
full (archive)	+0.0143	—	—
direct ($\gamma \cdot W_{\text{fast}}$ only)	+0.0148	+0.59	—
nocons (no write)	+0.0036	-5.53	+5.30
qonly ($\beta \cdot Q$ only; $n=4$)	+0.0094	$\approx$ nocons	$\approx$ nocons

direct $\approx$ full; removing consolidation drops the HE arm; the designed success-gated Q pathway (which is the only “selective” mechanism in the code, and is global-scalar by construction) adds nothing on top of no-consolidation. *Level: causal ($n=12$; $n=4$ for qonly).*

4.4 The allocation of the unselective write is functionally necessary — emergent selective learning

Energy-matched allocation shuffle (shufwrite): identical to direct except each sleep’s $\gamma \cdot W_{\text{fast}}$ increment is randomly permuted *within each matrix* before being added to W_{slow} . Total energy, magnitudes, rule and coefficients are identical; only *which coordinates* receive the write is destroyed.

HE arm, $n=12$: shufwrite mean +0.0042 vs direct +0.0148 $\rightarrow$ paired **t=4.86**; shufwrite $\approx$ nocons ($t=-0.38$) $\ll$ direct ($t=4.86$); direct $\approx$ full ($t=0.59$).

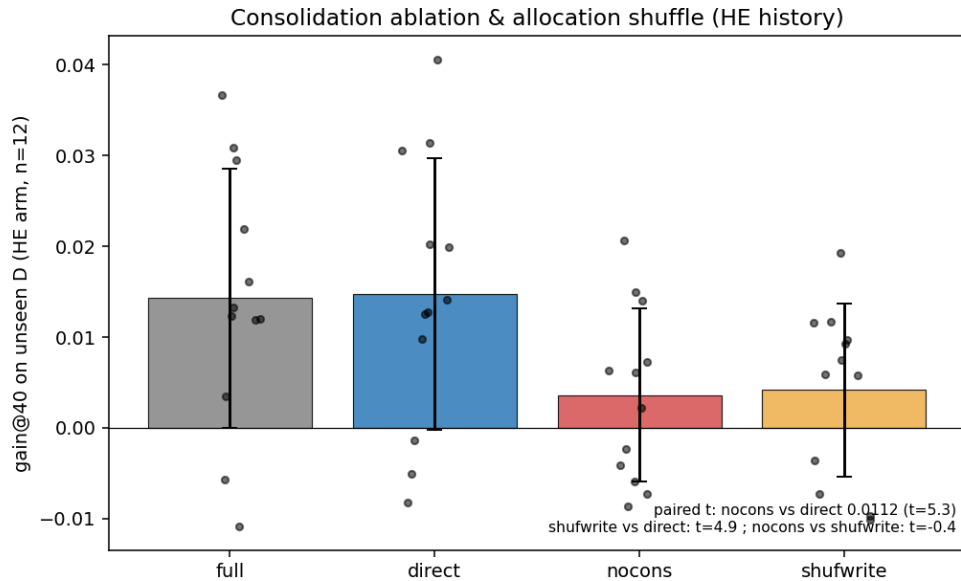


Figure 8: Consolidation ablation and allocation shuffle (HE arm, $n=12$).

Interpretation: the write rule is scalar-uniform and unselective *by rule*, yet its effect depends entirely on where the self-organized fast trace points it. The system therefore performs **allocation-level selection with no selection objective** — and the selection is not decorative: destroying it removes the effect. *Level: causal ($n=12$, single setting)*.

4.5 What does not happen (negative results, kept for honesty)

- The EH arm shows no such effect in any variant (all ≈ 0); the phenomenon is specific to the history/arm that carries the adaptation advantage.
- The designed “selective” mechanism (q , success) is numerically and causally inert (§4.3) — an explicit design coexisting with, but not causing, the emergent behavior.
- More experience does not make learning faster: matched-difficulty $p=0.17$, cross-domain 20-seed $p=0.82$, physics near-transfer $d=-0.17$.
- Macroscopic state geometry does not separate histories (norm/cosine at same-history cross-seed noise level; PCA PC1 $\approx 20\%$ variance).

4.6 Same-protocol retention and the sleep boundary (B1/B4)

End-of-history ppl on the three seen curriculum domains, relative to birth ppl (more negative = better retained; $n=12$, HE arm):

variant	rel. forgetting mean $\pm$ sd	paired vs direct (t)
direct	-0.0192 $\pm$ 0.0083	—
nocons	-0.0152 $\pm$ 0.0111	+2.03 (trend: direct $\geq$ nocons)
nosleep	-0.0049 $\pm$ 0.0079	+10.26

In the same runs the no-sleep individual is highest on the unseen-D probe (+0.0303 vs direct +0.0140, $t=+6.1$) but is far worse at retention — i.e. the sleep boundary (decay + moment reset) buys old-task memory at a modest forward cost, and the direct write is the only consolidation variant that keeps the forward benefit on top of it (uniformwrite/topwrite/shufwrite all $\approx$ nocons, $t\approx -4$ to -5.5).

4.7 Ten-task continual learning: emergent selective consolidation as a CL primitive (A2/B2)

One individual per seed learns 10 cross-domain tasks sequentially ($n=8$). Forward = normalised first-10-step loss slope; retention = end-of-history ppl on the two earliest tasks.

variant	forward (norm_slope10)	retention of t1–t2 (end ppl)
direct	-0.00110 $\pm$ 0.00125	38.27 $\pm$ 5.74
nocons	-0.00100 $\pm$ 0.00129	39.03 $\pm$ 6.01
nosleep	-0.00090 $\pm$ 0.00125	39.51 $\pm$ 6.07

Paired: retention direct vs nocons $t=-3.60$, direct vs nosleep $t=-10.27$; forward direct vs nocons $t=-1.64$ (not slower), direct vs nosleep $t=-3.93$ (faster). No forward plasticity collapse across the 10 tasks in any variant. Reading: a self-organized, unselective fast $\rightarrow$ slow write yields a genuine continual-learning primitive — protected memory plus maintained (not slower) forward adaptation, no replay buffer, importance gate or penalty term; its coordinate allocation carries the forward benefit (§4.4) and the sleep boundary carries the retention benefit (§4.6).

5. Discussion

What we mean by “emergent selective learning”. Not “the learner chooses experiences” and not “the learner computes importance”. Rather: a scalar-uniform consolidation rule, applied to a structured fast trace, produces per-coordinate differential retention that is *causally required* for the history effect. Selectivity here is a property of the

interaction between an unselective rule and a self-organized state, measurable only through interventions that destroy allocation (energy-matched shuffle).

What the finding does and does not imply for design. If allocation-selectivity emerges for free in minimal fast/slow learners, then “selectivity” need not be an added mechanism — and adding a *global* scalar gate (like success) does not create it. Obtaining *parameter-level* or *experience-level* selectivity would require genuinely new per-parameter/per-item signals and a channel with the same causal weight as the direct write (which currently dwarfs Q by $\sim 250\times$). This is directly relevant to plasticity-maintenance work (Dohare et al., 2024; Lyle et al., 2022; Nikishin et al., 2022) and to the design of “growing” learners.

Limits. One 6.59M backbone ($n=12$); raw EH/HE gap is seed-noisy (strong claims are within-seed paired); the directional-alignment result is correlational; retention is now measured in the same protocol (end-of-history ppl; 10-task sequences): the sleep boundary protects earlier tasks and direct write gives the best retention ($n=8-12$, one backbone, no tuned baselines in the ten-task run); q only was run at $n=4$; second-setting replication is $n=2$ and directional only.

Statement-level summary.

Claim	In code by design?	Observed	Causal evidence	Status
History effect localized to W_{fast}	—	yes	swap, $n=12$, controls	established
Explicit alignment operator	no	—	—	absent
Directional trace (predictive, module-aligned)	emergent	$r=0.87$	correlational only	emergent character
Designed selection via success-gated Q	global scalar only	$Q \approx 0.002$; $qonly \approx nocons$	inert	refuted here
Direct fast→slow	yes (unselective)	yes	$n=12$, $t \approx 5.3$	established

Claim	In code by design?	Observed	Causal evidence	Status
write carries the effect	scalar)			
Allocation of the write is necessary	emergent from W_{fast}	yes	n=12 energy-matched shuffle, $t \approx 4.9$	established (core)
Sleep boundary (decay+reset) protects earlier tasks	no-write keeps boundary	yes	n=12 retention, $t \approx 10$ vs nosleep	established
Direct write best retention among sleeping variants	emergent from W_{fast}	yes	n=12 & n=8 (10-task)	established
More experience → faster learning	—	null	—	not supported

6. Conclusion

Selectivity can emerge where none is designed. In a minimal fast/slow learner with no alignment objective, no per-parameter gate, and a scalar-uniform consolidation rule, the learner’s own gradient history shapes a fast trace whose coordinate structure determines where consolidation writes; keeping the total write energy identical but destroying that allocation removes the effect (n=12, paired $t \approx 4.9$). The architecture’s explicit, globally-gated “selective” pathway plays no measurable role. The lesson is twofold: emergent, allocation-level selective learning is real and causally potent in this setting; and designers should not assume that adding a selection *signal* (especially a global one) is what creates selective behavior—or that selection needs to be added at all.

Reproducibility

Code: <https://github.com/JayCRL/DLA> (paper + audit tooling under analysis/wfast_geom/, reports under report_output/). Protocol, hyperparameters, per-seed data and commit hashes are recorded for every claim. Mechanism-audit runs used a parity-validated harness (Apple M2 CPU; mean $|\Delta_{\text{ppl}}| \approx 0.24$ vs archived server runs). Scripts: stage55e_wfast_p0.py, validation_p0_controls.py, validation_baselines.py, stage6/7/8, analysis/wfast_geom/{geom, replay, p1, p1b, p1c, p2, p4_consolidation_geom, audit_cheap, audit_b3, audit_t10, unified_summary}.py.

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